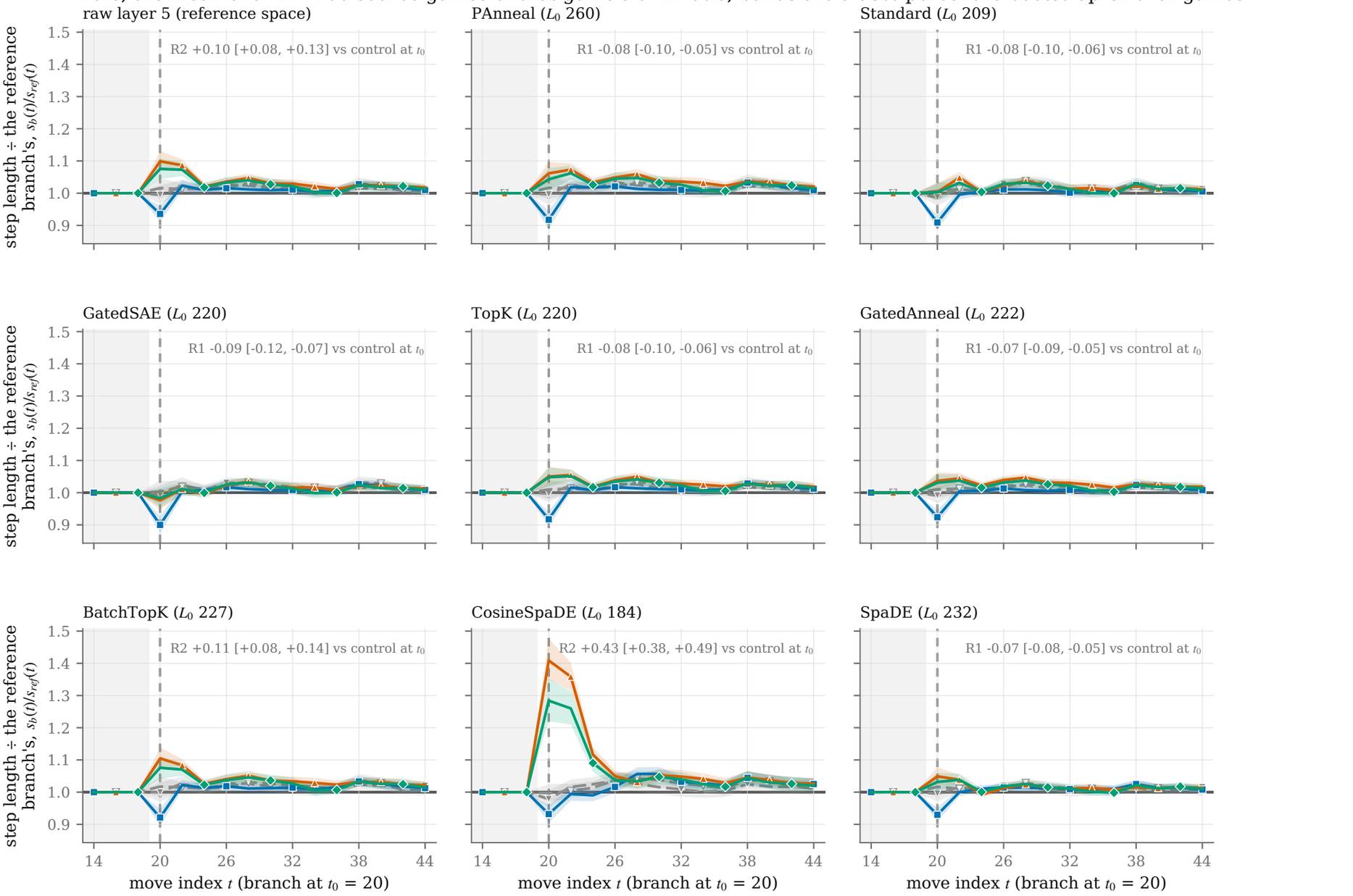
step length at the fork, as a multiple of the reference branch's at the same position — one curve per constraint rule, the mean over n = 100 source games of that game's own ratio; bands are a 95% percentile bootstrap CI over games



shared prefix: branches identical, ratio = 1 by construction
reference (the game as played), the line at 1
R1 placement (mean of 2)
R2 bracketing (mean of 3)
R3 turn (mean of 1)
95% percentile bootstrap CI of the mean over games
control: another legal move (null)
control 2: a third legal move (null)

one step spans 4 plies: positions are sampled every 2 plies at t_0's parity (fork_plane.positions), the fork family's convention. ../rule_curvature samples every ply instead, where the same layer-5 legal corpus steps 29.2 against 18.9 here (../theta_comparability) — the two conventions measure different quantities and must never share an axis.

the branch set is the rollout tree: after t_0 every branch — the reference included — samples its own continuation from OthelloGPT at temperature 1, 4 times each, instead of replaying the source game's. The ratio is formed within a rollout index — a branch's k-th draw against the reference's k-th, the one sharing its uniform stream — and averaged over the 4 afterwards, never ratio-of-means. Two branches now differ at every position after t_0 and not at t_0 alone, so the single-token attribution the spliced twin carries is gone (README).

the eight SAE cells are the l0_operating_point picks: one checkpoint per type, the nearest-L_0 member of its own grid to the shared target L_0 = 219.2, and each cell's own L_0 is in its title. The canonical figure seats each type at its own default sparsity instead, where a difference between two cells confounds the architecture with its L_0. Karvonen's five were trained on the raw residual stream and ours on scale·(x − mean); each is fed the activations it was trained on, so the two families' latents are not on one input scale (README).

within ÷ between-rule spread of the drawn ratio over $t \geq t_0$, in raw layer 5, per game over 100 games: median 0.896 [IQR 0.829–1.014], above 1 in 30% of them. Near 1 the three curves are a label rather than a finding; the nine spaces' values are in README.md.

the two grey curves are two draws of one null — a different legal token at t_0, measured exactly as the illegal branches are. Neither is a divisor here (the reference is), so the gap between them is the size of a difference that means nothing, and each panel's number is that rule minus control_legal on the same games, not two point estimates side by side.

the first 100 of the 1696 branch-complete games forking at t_0 = 20, ascending by game id: no game-level criterion enters this figure, and game 1235 — the argmax the single-game figure draws — is in it as one ordinary member of 100 (README).

R1 placement (2): square holds the mover's disc · square holds the opponent's disc
R2 bracketing (3): empty, touches no discs · empty, touches only the mover's discs · empty, touches opponent discs, no line closes
R3 turn (1): flips only for the opponent (turn confusion)